

1.09 Numerical Methods				
1.09a	Sign change methods		a) Be able to locate roots of $f(x) = 0$ by considering changes of sign of $f(x)$ in an interval of x on which $f(x)$ is sufficiently well-behaved. <i>Includes verifying the level of accuracy of an approximation by considering upper and lower bounds.</i>	MI1
1.09b			b) Understand how change of sign methods can fail. <i>e.g. when the curve $y = f(x)$ touches the x-axis or has a vertical asymptote.</i>	

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.
1.09c 1.09d 1.09e	Formal iterative methods		c) Be able to solve equations approximately using simple iterative methods, and be able to draw associated cobweb and staircase diagrams. d) Be able to solve equations using the Newton-Raphson method and other recurrence relations of the form $x_{n+1} = g(x_n)$. e) Understand and be able to show how such methods can fail. <i>In particular, learners should know that:</i> 1. <i>the iteration $x_{n+1} = g(x_n)$ converges to a root at $x = a$ if $g'(a) < 1$, and if x_1 is sufficiently close to a;</i> 2. <i>the Newton-Raphson method will fail if the initial value coincides with a stationary point.</i>	MI2
1.09f	Numerical integration		f) Understand and be able to use numerical integration of functions, including the use of the trapezium rule, and estimating the approximate area under a curve and the limits that it must lie between. <i>Learners will be expected to use the trapezium rule to estimate the area under a curve and to determine whether the trapezium rule gives an under- or over-estimate of the area under a curve.</i> <i>Learners will also be expected to use rectangles to estimate the area under a curve and to establish upper and lower bounds for a given integral. See also 1.08g.</i> <i>[Simpson's rule is excluded]</i>	MI3

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1.09g	Use numerical methods in context		g) Be able to use numerical methods to solve problems in context. <i>i.e. for solving problems in context which lead to equations which learners cannot solve analytically.</i>	MI4